# User Data Streams for Binance

## General information

- Subscribe via the WebSocket API using an API Key.
- Both SBE and JSON output are supported.
- Account events are pushed in **real-time**.
- All timestamps in JSON payloads are in **milliseconds by default**.
- Events may contain non-ASCII characters encoded in UTF-8 if you own or trade any assets or symbols whose names contain non-ASCII characters.

## User Data Stream Events

### Account Update

`outboundAccountPosition` is sent any time an account balance has changed and contains the assets that were possibly changed by the event that generated the balance change.

```
{
    "subscriptionId": 0,
    "event": {
        "e": "outboundAccountPosition",     // Event type
        "E": 1564034571105,                 // Event Time
        "u": 1564034571073,                 // Time of last account update
        // Balances Array
        "B": [
            {
                "a": "ETH",                 // Asset
                "f": "10000.000000",        // Free
                "l": "0.000000"             // Locked
            }
        ]
    }
}
```

### Balance Update

Balance Update occurs during the following:

- Deposits or withdrawals from the account
- Transfer of funds between accounts (e.g. Spot to Margin)

### Payload

```
{
    "subscriptionId": 0,
    "event": {
        "e": "balanceUpdate",      // Event Type
        "E": 1573200697110,        // Event Time
        "a": "BTC",                // Asset
        "d": "100.00000000",       // Balance Delta
        "T": 1573200697068         // Clear Time
    }
}
```

### Order Update

Orders are updated with the `executionReport` event.

### Payload:

```
{
    "subscriptionId": 0,
    "event": {
        "e": "executionReport",            // Event type
        "E": 1499405658658,                // Event time
        "s": "ETHBTC",                     // Symbol
        "c": "mUvoqJxFIILMdfAW5iGSOW",     // Client order ID
        "S": "BUY",                        // Side
        "o": "LIMIT",                      // Order type
        "f": "GTC",                        // Time in force
        "q": "1.00000000",                 // Order quantity
        "p": "0.10264410",                 // Order price
        "P": "0.00000000",                 // Stop price
        "F": "0.00000000",                 // Iceberg quantity
        "g": -1,                           // OrderListId
        "C": "",                           // Original client order ID; This is the ID of the
order being canceled
        "x": "NEW",                        // Current execution type
        "X": "NEW",                        // Current order status
        "r": "NONE",                       // Order reject reason; Please see Order Reject
Reason (below) for more information.
        "i": 4293153,                      // Order ID
        "l": "0.00000000",                 // Last executed quantity
        "z": "0.00000000",                 // Cumulative filled quantity
        "L": "0.00000000",                 // Last executed price
        "n": "0",                          // Commission amount
        "N": null,                         // Commission asset
        "T": 1499405658657,                // Transaction time
        "t": -1,                           // Trade ID
        "v": 3,                            // Prevented Match Id; This is only visible if the
order expired due to STP
        "I": 8641984,                      // Execution Id
        "w": true,                         // Is the order on the book?
        "m": false,                        // Is this trade the maker side?
        "M": false,                        // Ignore
```

```
        "O": 1499405658657,                // Order creation time
        "Z": "0.00000000",                 // Cumulative quote asset transacted quantity
        "Y": "0.00000000",                 // Last quote asset transacted quantity (i.e.
lastPrice * lastQty)
        "Q": "0.00000000",                 // Quote Order Quantity
        "W": 1499405658657,                // Working Time; This is only visible if the order
has been placed on the book.
        "V": "NONE"                        // SelfTradePreventionMode
    }
}
```

**Note:** Average price can be found by doing `Z` divided by `z`.

### Conditional Fields in Execution Report

These are fields that appear in the payload only if certain conditions are met.

For additional information on these parameters, please refer to the Spot Glossary.

| Field | Name | Description | Examples |
|---|---|---|---|
| `d` | Trailing Delta | Appears only for trailing stop orders. | `"d": 4` |
| `D` | Trailing Time | | `"D": 1668680518494` |
| `j` | Strategy Id | Appears only if the `strategyId` parameter was provided upon order placement. | `"j": 1` |
| `J` | Strategy Type | Appears only if the `strategyType` parameter was provided upon order placement. | `"J": 1000000` |
| `v` | Prevented Match Id | Appears only for orders that expired due to STP. | `"v": 3` |
| `A` | Prevented Quantity | | `"A":"3.000000"` |
| `B` | Last Prevented Quantity | | `"B":"3.000000"` |
| `u` | Trade Group Id | | `"u":1` |
| `U` | Counter Order Id | | `"U":37` |

| | | | |
|---|---|---|---|
| `Cs` | Counter Symbol | | `"Cs": "BTCUSDT"` |
| `pl` | Prevented Execution Quantity | | `"pl":"2.123456"` |
| `pL` | Prevented Execution Price | | `"pL":"0.10000001"` |
| `pY` | Prevented Execution Quote Qty | | `"pY":"0.21234562"` |
| `W` | Working Time | Appears when the order is working on the book | `"W": 1668683798379` |
| `b` | Match Type | Appears for orders that have allocations | `"b":"ONE_PARTY_TRADE_REPORT"` |
| `a` | Allocation ID | | `"a":1234` |
| `k` | Working Floor | Appears for orders that potentially have allocations | `"k":"SOR"` |
| `uS` | UsedSor | Appears for orders that used SOR | `"uS":true` |
| `gP` | Pegged Price Type | Appears only for Pegged Orders | `"gP": "PRIMARY_PEG"` |
| `gOT` | Pegged offset Type | | `"gOT": "PRICE_LEVEL"` |
| `gOV` | Pegged Offset Value | | `"gOV": 5` |
| `gp` | Pegged Price | | `"gp": "1.00000000"` |
| `eR` | Expiry Reason | Appears when the order has expired. | `"eR": “INSUFFICIENT_LIQUIDITY”` |

## Order Reject Reason

For additional details, look up the Error Message in the Errors documentation.

| Rejection Reason (`r`) | Error Message |
| --- | --- |
| `NONE` | N/A (i.e. The order was not rejected.) |
| `INSUFFICIENT_BALANCES` | "Account has insufficient balance for requested action." |
| `STOP_PRICE_WOULD_TRIGGER_IMMEDIATELY` | "Order would trigger immediately." |
| `WOULD_MATCH_IMMEDIATELY` | "Order would immediately match and take." |
| `OCO_BAD_PRICES` | "The relationship of the prices for the orders is not correct." |

If the order is an order list, an event named `ListStatus` will be sent in addition to the `executionReport` event.

### Payload

```
{
    "subscriptionId": 0,
    "event": {
        "e": "listStatus",                      // Event Type
        "E": 1564035303637,                     // Event Time
        "s": "ETHBTC",                          // Symbol
        "g": 2,                                 // OrderListId
        "c": "OCO",                             // Contingency Type
        "l": "EXEC_STARTED",                    // List Status Type
        "L": "EXECUTING",                       // List Order Status
        "r": "NONE",                            // List Reject Reason
        "C": "F4QN4G8DlFATFlIUQ0cjdD",          // List Client Order ID
        "T": 1564035303625,                     // Transaction Time
        // An array of objects
        "O": [
            {
                "s": "ETHBTC",                  // Symbol
                "i": 17,                        // OrderId
                "c": "AJYsMjErWJesZvqlJCTUgL"   // ClientOrderId
            },
            {
                "s": "ETHBTC",
                "i": 18,
                "c": "bfYPSQdLoqAJeNrOr9adzq"
            }
        ]
    }
}
```

### Order Expiry Reason

| Expiry Reason (`eR`) | Explanation |
| --- | --- |
| `REJECTED` | A contingent order or an order that was part of an OTO was rejected by the matching engine when trying to place it on the order book. Common reasons are lack of funds and rejection by filters. |
| `EXCHANGE_CANCELED` | The order was canceled by Binance. |
| `OCO_TRIGGER` | An order that was part of an OCO pair was canceled because the other order of the pair started working or the entire OCO expired. |
| `OTO_PHASE_ONE_EXPIRED` | The working order of the order list expired, thus expiring the entire order list. |
| `UNFILLED_IOC_QUANTITY_EXPIRED` | The IOC order was not fully filled and thus expired. |
| `UNFILLED_FOK_ORDER_EXPIRED` | The FOK order was not fully filled and thus expired. |
| `INSUFFICIENT_LIQUIDITY` | There were not enough orders in the order book to match with this order. |
| `EXECUTION_RULE_PRICE_RANGE_EXCEEDED` | The order attempted to trade at a price that would not meet the Price Range Execution Rule. |

Check the Enums page for more relevant enum definitions.

## Event Stream Terminated

`eventStreamTerminated` is sent when:

- A listen token subscription expires due to token expiration.
- A logon subscription ends after sending `session.logout` method.
- The subscription is stopped via the `userDataStream.unsubscribe` method.

**Payload:**

```
{
    "subscriptionId": 0,
    "event": {
        "e": "eventStreamTerminated",      // Event Type
        "E": 1728973001334                 // Event Time
    }
}
```

## External Lock Update

`externalLockUpdate` is sent when part of your spot wallet balance is locked/unlocked by an external system, for example when used as margin collateral.

**Payload:**

```
{
    "subscriptionId": 0,
    "event": {
        "e": "externalLockUpdate",     // Event Type
        "E": 1581557507324,            // Event Time
        "a": "NEO",                    // Asset
        "d": "10.00000000",            // Delta
        "T": 1581557507268             // Transaction Time
    }
}
```